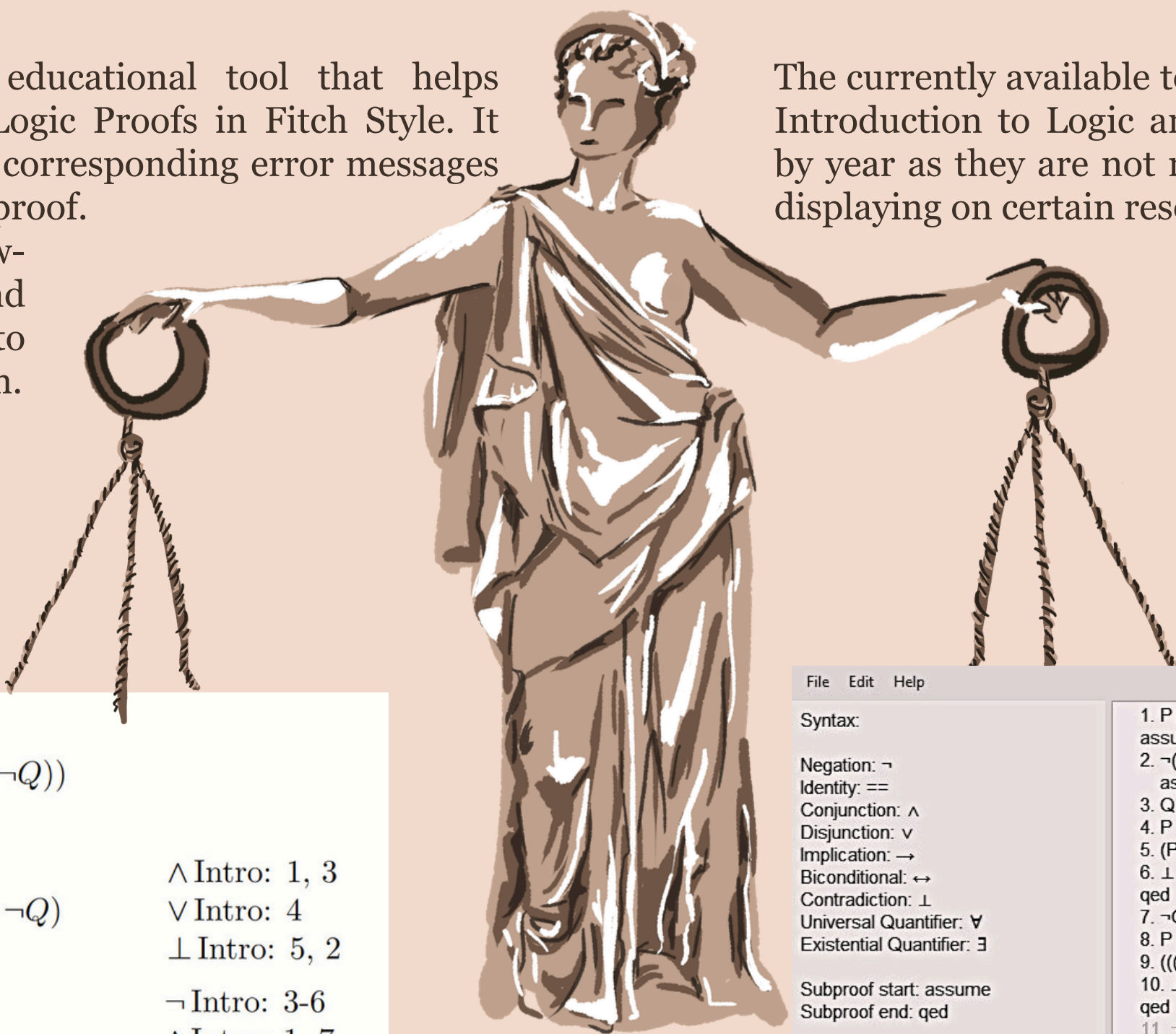


1) ABSTRACT

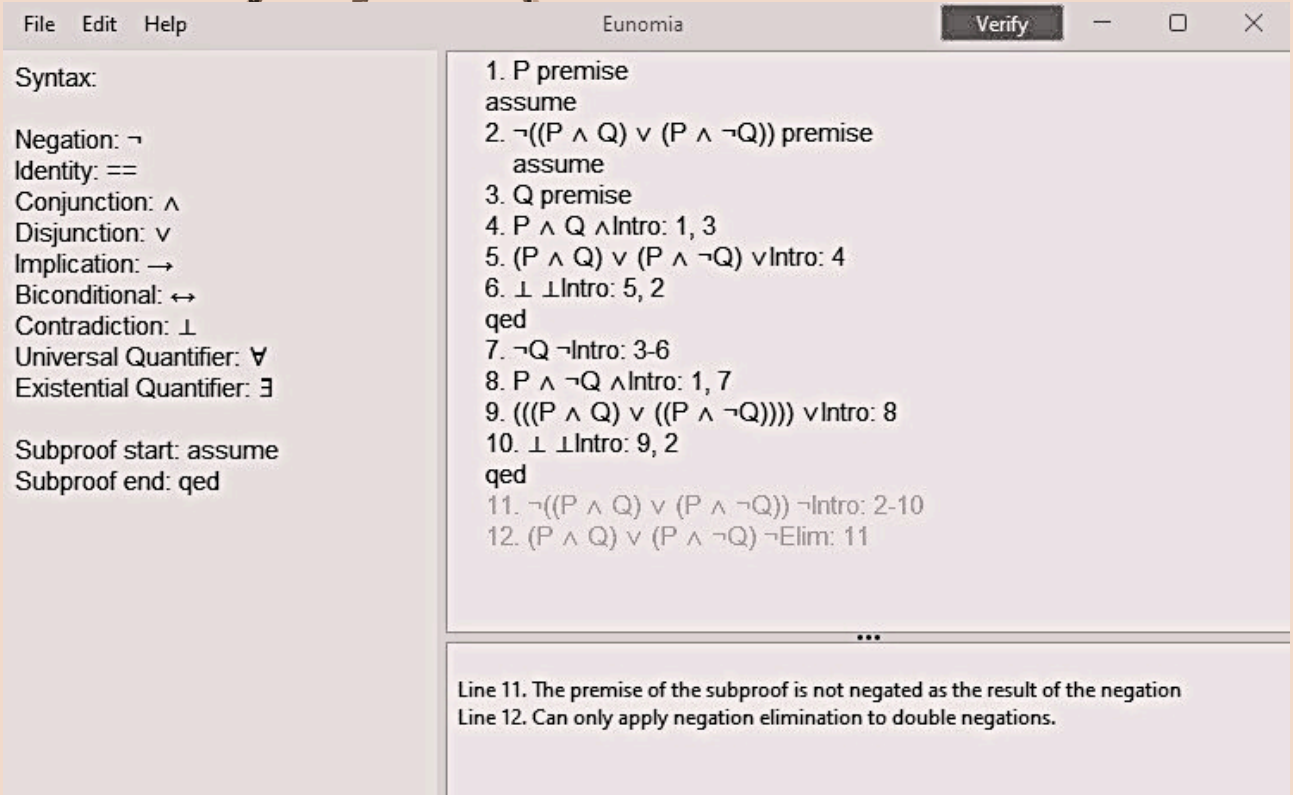
Eunomia represents an educational tool that helps students in constructing Logic Proofs in Fitch Style. It achieves this by providing corresponding error messages for each wrong step of the proof. It is built on a Model-View-Controller architecture and it is thoroughly tested to guarantee correct execution.

2) RATIONALE

The currently available tools at the university for teaching Introduction to Logic are becoming more outdated year by year as they are not maintained. The issues related to displaying on certain resolutions, hard accessibility due to the need to access the university Virtual Machine, the cluttered UI and not consistent style with the course standards burden the students unnecessarily.



1. P	
2. $\neg((P \wedge Q) \vee (P \wedge \neg Q))$	
3. Q	
4. $P \wedge Q$	$\wedge$ Intro: 1, 3
5. $(P \wedge Q) \vee (P \wedge \neg Q)$	$\vee$ Intro: 4
6. $\bot$	$\bot$ Intro: 5, 2
7. $\neg Q$	$\neg$ Intro: 3-6
8. $P \wedge \neg Q$	$\wedge$ Intro: 1, 7
9. $(P \wedge Q) \vee (P \wedge \neg Q)$	$\vee$ Intro: 8
10. $\bot$	$\bot$ Intro: 9, 2
11. $\neg\neg((P \wedge Q) \vee (P \wedge \neg Q))$	$\neg$ Intro: 2-10
12. $(P \wedge Q) \vee (P \wedge \neg Q)$	$\neg$ Elim: 11



3) COMPILING THE PROOF

The software is built on the MVC (Model-View-Controller) architecture. The model is responsible for the logic behind the system while the view is responsible for the UI.

Compilation of the proof happens in the model. ANTLR (ANother Tool for Language Recognition) was used to generate a parser that accepts the language of a proof in Fitch Style.

- i) First, we defined a grammar that represents the format of a proof and ANTLR was used to generate a parser for it.
- ii) Then, the parser saves all the information of the proof in a tree data structure which is navigable using a Visitor Design Pattern.
- iii) Finally, using the visitor we go line by line through the proof and decide if it follows from the previous lines. We store all the required information and all the errors.

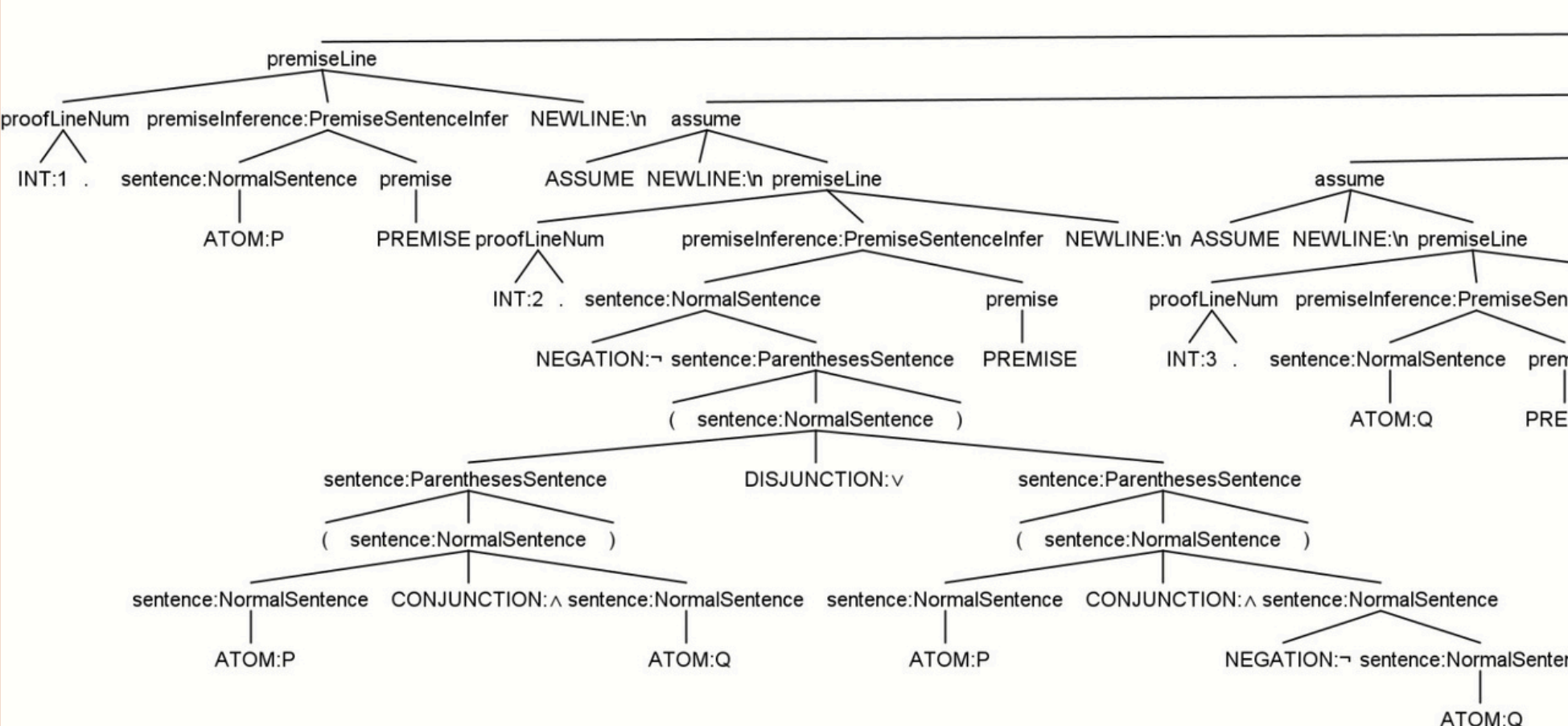
EUNOMIA

4) USER INTERFACE The UI is based on the layout of famous code editors, like Visual Studio Code. This design choice had the intention of bringing familiarity and facilitating the use of the software. There are three main panels.

- i) An information panel bringing information related to logic proofs or the utilization of the software.
- ii) The write panel where the proof is situated.
- iii) The error panel where all the error messages are displayed.

5) LEARNINGS

This project allowed me to delve more into topics related to Formal Grammars, Compiler Construction, Parsing and Software Engineering practices. Furthermore, during development I had to get accustomed to using ANTLR which is one of the most advanced tools for language recognition and parser generation.



6) CONCLUSION

The tool will be used in the 2023 iteration of the course Introduction to Logic. This will also be a great opportunity for user feedback and user acceptance tests. Because of the MVC architecture, the project allows for easy maintainability in the future in the case of other developers wishing to contribute. The software will also be made open source.